

8. [Maximum mark: 7]

(a) Find the first two non-zero terms in the Maclaurin series of

(i) $\sin(x^2)$;

(ii) $\sin^2(x^2)$.

[5]

(b) Hence, or otherwise, find the first two non-zero terms in the Maclaurin series of $4x \sin(x^2) \cos(x^2)$.

[2]

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